Quantum-INS Benchmark Workflow

How the Claude Scholar research lifecycle, the 10-agent build/analysis pipeline, and the project's skills/commands fit together — worked through one real project end to end.

Project: extend the single-model result in *Agent division.pdf* and *NS_APS_MM_2026.pptx* (“Benchmarking quantum simulation with neutron scattering experiments,” arXiv:2603.15608 — $KCuF_3$ only, APS March Meeting 2026) into a four-model benchmark suite — XX chain, $KCuF_3$, two-soliton, $CsCoX_3$ — comparing IBM hardware, noisy-MPS simulation, and classical DMRG against experimental INS spectra.

1

Research Ideation

SKILL / AGENT	`research-ideation` skill + `literature-reviewer` agent + Zotero MCP
COMMAND	`/research-init`
EXAMPLE	`literature-reviewer` scans the Zotero “INS-Quantum-Simulation” collection and flags that the published $KCuF_3$ result has no counterpart for a gapless chain (XX), a soliton lattice, or a frustrated triangular lattice ($CsCoX_3$). `research-ideation` runs 5W1H on that gap.
OUTCOME	`/plan/2026-10-09-four-model-benchmark-ideation.md` — research question (“does DSF-reconstruction fidelity degrade systematically with magnon bandwidth / gap structure across hardware backends?”), 4-model justification table, Zotero collection link.

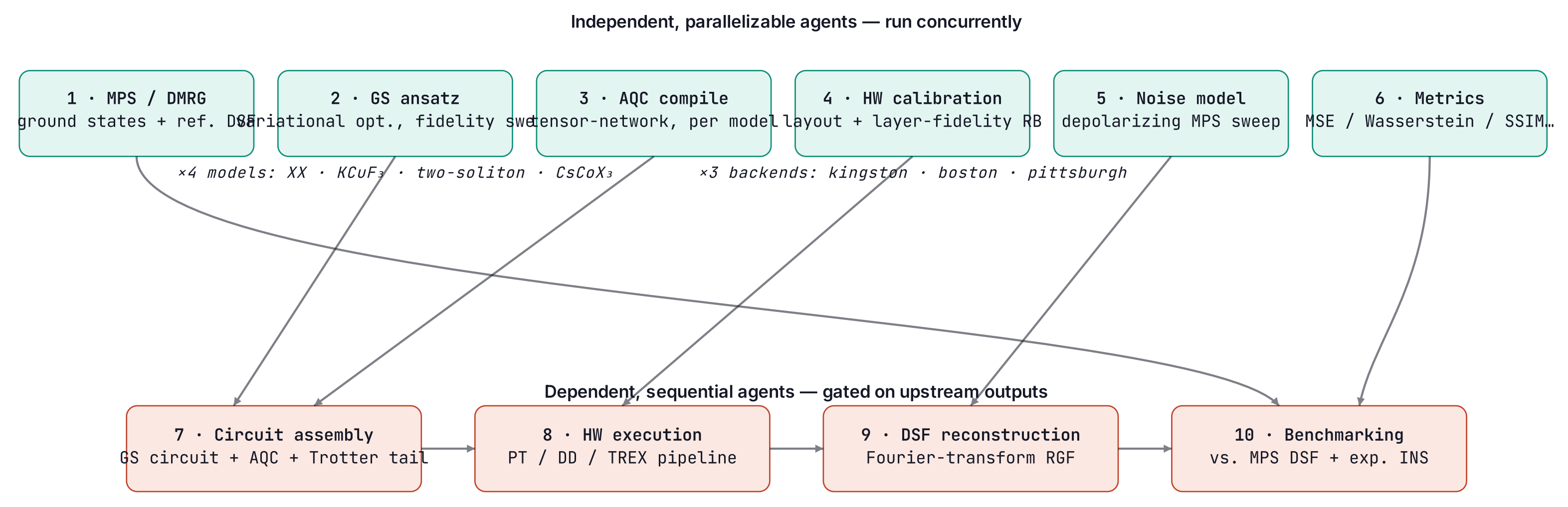
2

ML / Simulation Development

SKILL / AGENT	`architecture-design` skill + `dev-planner` + `code-reviewer` agent
COMMAND	`/plan` → `/tdd` → `/commit`
EXAMPLE	`dev-planner` turns *Agent division.pdf*'s 10-agent split into an implementation plan: a Hydra `model=xx|kcuf3|two_soliton|csocox3` registry (factory pattern) and a DAG scheduler gating agents 7–10 on upstream futures. `code-reviewer` checks the metrics agent (6) is unit-tested against dummy spectra before real data exists, per the PDF's own note.
OUTCOME	`src/model_module/{xx,kcuf3,two_soliton,csocox3}/` scaffolding, `run/pipeline/agent_dag.py` scheduler, a passing test suite for MSE / Wasserstein / SSIM / nQFI / two-tangle against synthetic spectra, commit `feat(pipeline): add 10-agent benchmarking DAG scheduler`.

▸ builds the pipeline shown in Figure 2, below

Figure 2 — the 10-agent build/analysis DAG


Independent, parallelizable agents — run concurrently
1 · MPS / DMRG
ground states + ref. DSF
2 · GS ansatz
variational opt., fidelity sweep
3 · AQC compile
tensor-network, per model
4 · HW calibration
layout + layer-fidelity RB
5 · Noise model
depolarizing MPS sweep
6 · Metrics
MSE / Wasserstein / SSIM…
×4 models: XX · KCuF₃ · two-soliton · CsCoX₃
×3 backends: kingston · boston · pittsburgh
Dependent, sequential agents — gated on upstream outputs
7 · Circuit assembly
GS circuit + AQC + Trotter tail
8 · HW execution
PT / DD / TREX pipeline
9 · DSF reconstruction
Fourier-transform RGF
10 · Benchmarking
vs. MPS DSF + exp. INS

Six agents (1–6) need nothing but the four model definitions and run in parallel — agents 1–3 fan out ×4 models, agent 4 fans out ×3 hardware backends. Agent 7 can only start once 2 and 3 finish; 8 needs 7 and 4; 9 needs 8 and 5; 10 — the benchmarking step — waits on 9 plus the two agents (1 and 6) that were ready earliest but are consumed last.

3

Experiment Analysis

SKILL	`results-analysis` + `results-report` skills
COMMAND	`/analyze-results`
EXAMPLE	Once agent 10 emits per-model, per-backend tables, `results-analysis` runs significance tests across the $\chi=16\rightarrow256$ bond-dimension convergence sweep and the $7.5\times10^{-3}\rightarrow1\times10^{-3}$ error-rate sweep. The published $KCuF_3$ baseline (MSE≈2×10^{-3}, Wasserstein≈1.5×10^{-2}, SSIM≈0.65) becomes the reference row the other three models are judged against.
OUTCOME	`results/2026-xx_four-model-benchmark-report.md` with a metric table per model × backend, a convergence plot, and a finding such as “$CsCoX_3$ (frustrated, largest gap) shows 2.3× higher Wasserstein distance on hardware vs. noisy-MPS — crosstalk outside the depolarizing model.”

4

Paper Writing

SKILL / AGENT	`ml-paper-writing` skill + `paper-miner` agent + `citation-verification`
COMMAND	`/mine-writing-patterns`
EXAMPLE	`paper-miner` extracts structure and tone from comparable PRX Quantum / Nature Physics benchmarking papers. `ml-paper-writing` drafts the extended manuscript — same narrative arc as the APS talk (`NS_APS_MM_2026.pptx`), now with four Results panels instead of one, and a Methods subsection mapped 1:1 onto agents 1–10.
OUTCOME	`paper/draft_v1.tex` — expanded Results/Methods, citation-verified INS and DMRG references.

5

Self-Review

SKILL	`paper-self-review` (6-item checklist)
EXAMPLE	Checklist flags three gaps: no citation justifying the depolarizing-channel assumption (agent 5), missing gate count for the two-soliton AQC compilation (agent 3), and no recorded seed for the ground-state ansatz optimization (agent 2).
OUTCOME	`self-review/checklist-v1.md` — 3 open items, each tagged to its originating agent, routed back into Stage 2 before submission.

6

Submission & Rebuttal

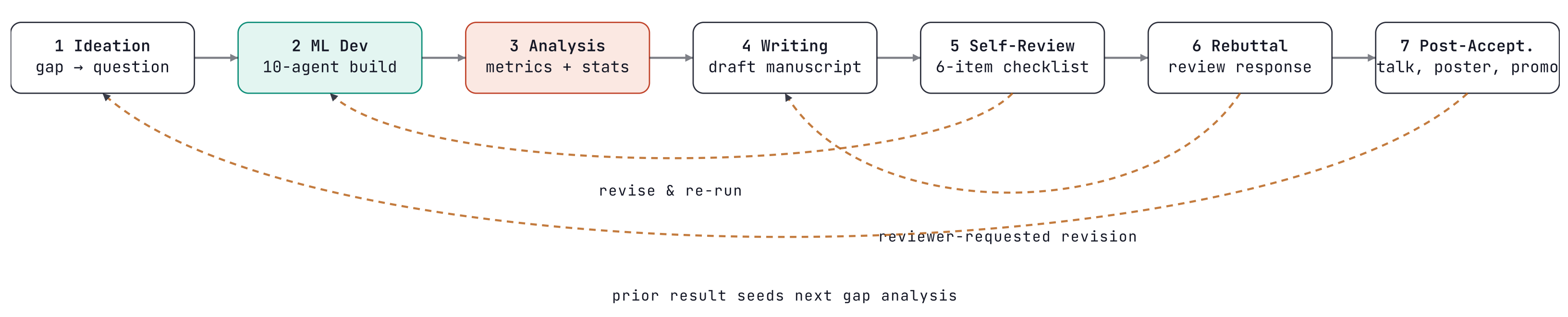
SKILL / AGENT	`review-response` skill + `rebuttal-writer` agent
COMMAND	`/rebuttal`
EXAMPLE	Reviewer: “How do you know the depolarizing noise model matches the real device?” `rebuttal-writer` answers by citing agent 4's layer-fidelity RB data as the empirical cross-check, and requests one new supplementary figure from Stage 2/3.
OUTCOME	`rebuttal/response-to-reviewer-2.md` — point-by-point table + one new supplementary comparison plot.

7

Post-Acceptance

SKILL	`post-acceptance` skill
COMMAND	`/presentation` · `/poster` · `/promote`
EXAMPLE	`NS_APS_MM_2026.pptx` (the single-model talk) is superseded by a four-model deck; `/poster` turns the same content into a conference poster; `/promote` drafts a two-post thread on the cross-model result.
OUTCOME	`presentation/NS_APS_MM_2027_v2.pptx`, `poster/four-model-benchmark-poster.pdf`, `promote/thread-draft.md`.

Figure 1 — the seven-stage lifecycle, with its feedback loops


1 Ideation
gap → question
2 ML Dev
10-agent build
3 Analysis
metrics + stats
4 Writing
draft manuscript
5 Self-Review
6-item checklist
6 Rebuttal
review response
7 Post-Accept.
talk, poster, promo
revise & re-run
reviewer-requested revision
prior result seeds next gap analysis

Stages 2 and 3 run the 10-agent DAG of Figure 2. Three dashed loops are the feedback paths that actually recur in this project: a self-review finding sends work back to ML Dev, a reviewer request sends work back to Paper Writing, and — the loop already visible in the source material — the accepted single-model result ($KCuF_3$, arXiv:2603.15608) is what seeded this project's own Stage-1 gap analysis.